Aymen Akdim



Embedded Systems Master's Student - PFE Candidate (Embedded & Automotive)

+212 6 98 51 12 13 | aymen.akdim@uit.ac.ma | Kénitra, Morocco | GitHub: github.com/aymenakdim

PROFILE

Master's student in Embedded Electronics (Ibn Tofail University, Faculty of Sciences) with a Bachelor's degree in Electrical Engineering, Instrumentation & Supervision. Passionate about automotive embedded software, I develop in C/C++ on microcontrollers and Raspberry Pi, and I built an ECU network simulator communicating over a CAN bus with OBD-II and UDS diagnostics. Used to testing, debugging and documenting my own developments, I am looking for a **6-month end-of-studies internship (PFE) starting between January and February 2027 (flexible dates)** in embedded systems, to contribute to the development, integration and validation of automotive systems.

KEY SKILLS

Embedded programming: C (C11), C++, Python, MATLAB / Simulink, VHDL / Verilog
Microcontrollers & architectures: Arduino, ESP32, 8051, Raspberry Pi 4, FPGA
Automotive & protocols: CAN (encoding/decoding), ECU, OBD-II (DTC), UDS (ISO 14229), ISO-TP, ADAS, I2C, MQTT
Embedded vision & AI: CNN, TensorFlow Lite, OpenCV, image processing
Testing, validation & tools: Unit & non-regression testing, fault injection, pytest, CMake, Git / GitHub, Proteus, Grafana
Other: Flask, Flutter, Firebase RTDB, Node-RED, TIA Portal & HMI, MS Office

PROJECTS

Automotive ECU Network & Diagnostic Simulator *Personal project*
C11 · GCC (MSYS2/MinGW-w64) · CMake · Git/GitHub · Python/pytest · C unit-test framework
- Simulator in C of an automotive ECU network (engine ECU, dashboard) exchanging 8-byte CAN frames over a simulated bus.
- Engine ECU: sensor validation, 10 Hz control cycle, OBD-II DTC management; dashboard timeout (stale data) monitoring.
- Fault injection (dropped frames, bit-flips, invalid IDs/DLC) and UDS (ISO 14229) diagnostics over ISO-TP.
- Custom C unit-test framework (450+ assertions) with non-regression tests.

Traffic Sign Detection (ADAS) *Personal project*
Raspberry Pi 4 Model B · Webcam · CNN (GTSRB) · TensorFlow Lite · OpenCV · Flask
- CNN trained on GTSRB, converted to TensorFlow Lite and deployed on a Raspberry Pi 4 with a webcam for real-time detection.
- OpenCV image processing and Flask web interface showing the video stream and results; validated under varied conditions.

Smart Soap Dispenser (IoT) *Personal project*
Microcontroller · Firebase Realtime Database · Flutter · Web
- Full integration: sensors/actuators, microcontroller, cloud, web and mobile application.
- Real-time supervision (distribution data, equipment status) via a Flutter app and a Firebase-synced website.

FPGA Simulation and Programming *Academic project*
VHDL · Verilog
- Simulation of digital circuits on an FPGA board; design and simulation of a simplified RISC microprocessor.

EXPERIENCE

Electrical Maintenance Intern – OCP *1 month · 2024*
- Diagnosis and corrective maintenance of electric motors (disassembly, replacement of bearings, windings and brushes).
- Low- and high-voltage electrical tests (insulation, dielectric strength, load tests): measurement rigour and failure analysis in an industrial environment.

EDUCATION

Master's in Embedded Electronics *2025 – 2027*
Ibn Tofail University, Faculty of Sciences, Kénitra · ongoing

Bachelor's in Electrical Engineering, Instrumentation and Supervision of Smart Systems *2024 – 2025*
EST, Kénitra

Diploma in Electromechanics of Automated Systems *2022 – 2024*
ISTA, Khouribga

LANGUAGES & INTERESTS

Arabic (native) · **French** (intermediate) · **English** (upper-intermediate)
Interests: Strategy video games, Volleyball